Applied math notes

Lecture 1: The Galerkin method

Solutions to the exercises

Exercise 1: Analytical solution and error norm for problem 2

Problem 2: $-u'' = 1$ on $(0,1)$, $u(0) = 0$, $u'(1) = 0$.

1. Integrating $u'' = -1$ twice gives $u' = -x + C_1$ and $u = -\frac{1}{2}x^2 + C_1 x + C_2$. The boundary conditions give $u'(1) = -1 + C_1 = 0$, so $C_1 = 1$, and $u(0) = C_2 = 0$:

$$u(x) = x - \tfrac{1}{2}x^2.$$

2. The FEniCS script below (20 elements, as in the notebook) solves the problem with Lagrange elements of degree 1, 2 and 3. For degree 1 the nodal values of u_h agree with u up to round-off, $\max_i |u(x_i) - u_h(x_i)| = 1.1 \cdot 10^{-14}$: **the FEM solution is exact at the nodes.**

Reason: the error $e = u - u_h$ satisfies $a(e, v_h) = 0$ for all $v_h \in V_h$ (Galerkin orthogonality). With the hat function $v_h = h_i$, whose derivative is $\pm 1/h$ on the two elements around node i,

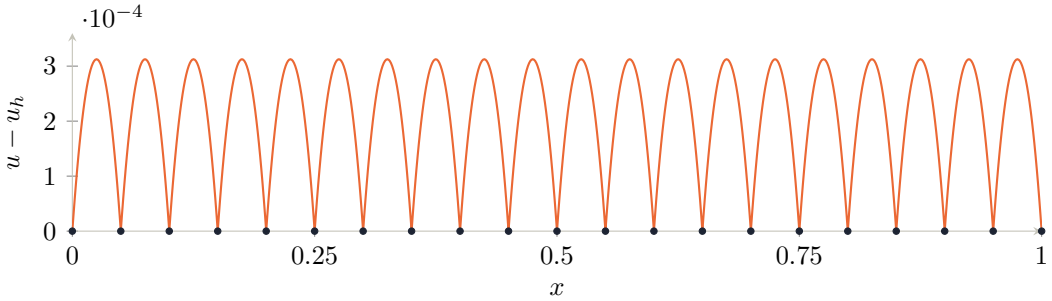
$$\int_0^1 e'\, h_i'\, dx = \frac{1}{h}(e(x_i) - e(x_{i-1})) - \frac{1}{h}(e(x_{i+1}) - e(x_i)) = 0,$$

i.e. $-e_{i-1} + 2e_i - e_{i+1} = 0$ for the nodal errors; at the Neumann end $e_N - e_{N-1} = 0$, and $e_0 = 0$. Hence $e_i = 0$ at all nodes. (This holds in 1D for this operator, not in general.)

Between the nodes u_h is therefore the linear interpolant of u. On an element $[x_{i-1}, x_i]$ the error vanishes at both ends and has $e'' = u'' = -1$, so

$$e(x) = \tfrac{1}{2}\,(x - x_{i-1})\,(x_i - x), \qquad \max e = \frac{h^2}{8} = 3.1 \cdot 10^{-4} \quad (h = 1/20),$$

a parabola on every element:



3. With $s = x - x_{i-1}$ and $\int_0^h s^2(h-s)^2\, ds = h^5/30$, summed over the $N = 1/h$ elements,

$$\|u - u_h\|_{L^2}^2 = N \cdot \frac{1}{4} \cdot \frac{h^5}{30} = \frac{h^4}{120}, \qquad \|u - u_h\|_{L^2} = \frac{h^2}{\sqrt{120}} = 2.282 \cdot 10^{-4} \quad (h = 1/20).$$

FEniCS gives the same value with `errornorm(u_exact, uh, 'L2')`. The error is $O(h^2)$: halving h quarters it.

4. Results of the script:

degree	max. nodal error	$\|u - u_h\|_{L^2}$
1	$1.1 \cdot 10^{-14}$	$2.282 \cdot 10^{-4}$
2	$1.6 \cdot 10^{-13}$	$1.1 \cdot 10^{-13}$
3	$7.1 \cdot 10^{-13}$	$5.1 \cdot 10^{-13}$

From degree 2 on the error is round-off. The exact solution $u = x - \frac{1}{2}x^2$ is a quadratic polynomial, so it lies in V_h for degree ≥ 2, and the Galerkin method reproduces any solution that lies in V_h: $u_h = u$.

```
# Lecture 1, exercise 1: -u'' = 1 on (0, 1), u(0) = 0, u'(1) = 0, exact solution u = x - x
    ↪ ^2/2
from fenics import *

set_log_level(LogLevel.WARNING)

mesh = IntervalMesh(20, 0.0, 1.0)
u_exact = Expression('x[0] - 0.5*x[0]*x[0]', degree=6)

def left(x, on_boundary):
    return on_boundary and near(x[0], 0.0)

for degree in [1, 2, 3]:
    V  = FunctionSpace(mesh, 'Lagrange', degree)
    u  = TrialFunction(V)
    v  = TestFunction(V)
    uh = Function(V)
    solve(inner(grad(u), grad(v))*dx == Constant(1.0)*v*dx, uh, DirichletBC(V, Constant(0.0),
    ↪  left))

    nodes = mesh.coordinates()[:, 0]
    nodal = max(abs(uh(x) - u_exact(x)) for x in nodes)
    print('degree %d: max nodal error %.1e, L2 error %.3e' % (degree, nodal, errornorm(
    ↪ u_exact, uh, 'L2')))
```

Exercise 2: Biharmonic equation (theory)

$Au = u'''' = f$ on $\Omega = (0, 1)$.

1. The highest derivative is the fourth: $O(A) = 4 = 2s$, so $s = 2$ and $m = s - 1 = 1$.

2. Integral form and two integrations by parts:

$$(Au, v) = \int_0^1 u'''' \, v \, dx = \left[u''' \, v\right]_0^1 - \int_0^1 u''' \, v' \, dx = \left[u''' \, v - u'' \, v'\right]_0^1 + \int_0^1 u'' \, v'' \, dx.$$

Variational formulation:

$$\int_0^1 u'' \, v'' \, dx + \left[u''' \, v - u'' \, v'\right]_0^1 = \int_0^1 f \, v \, dx \qquad \text{for all } v.$$

Boundary terms at $x = 0$ and $x = 1$: $u''' \, v$ (shear force times deflection) and $u'' \, v'$ (bending moment times slope).

3. With $m = 1$:

prescribed	c	type	condition on v
u (deflection)	0	Dirichlet (essential)	$v = 0$ there
u' (slope)	1	Dirichlet (essential)	$v' = 0$ there
u'' (bending moment)	2	Neumann (natural)	none
u''' (shear force)	3	Neumann (natural)	none

Where u or u' is prescribed, $v = 0$ or $v' = 0$ removes the corresponding boundary term. Where u'' or u''' is prescribed, the boundary term contains known values and moves to the right-hand side as a load.

4. The Hermite functions satisfy $h_1(0) = 1$, $h_2'(0) = 1$, $h_3(1) = 1$, $h_4'(1) = 1$ and all other values and slopes at $x = 0, 1$ are zero; e.g. $h_1' = -6x + 6x^2$ gives $h_1'(0) = h_1'(1) = 0$ and $h_1(1) = 1 - 3 + 2 = 0$. Hence $U_1 = u_h(0)$, $U_2 = u_h'(0)$, $U_3 = u_h(1)$, $U_4 = u_h'(1)$.

a) The clamping gives $U_1 = U_2 = 0$, and admissible test functions satisfy $v(0) = v'(0) = 0$: these are h_3 and h_4 ($h_1(0) = 1$, $h_2'(0) = 1$). So rows 3 and 4 are needed (test functions h_3, h_4), and only columns 3 and 4 (columns 1 and 2 multiply $U_1 = U_2 = 0$).

b) With $h_3'' = 6 - 12x$ and $h_4'' = -2 + 6x$,

$$K_{33} = \int_0^1 (6 - 12x)^2 \, dx = 12, \qquad K_{44} = \int_0^1 (-2 + 6x)^2 \, dx = 4,$$
$$K_{34} = K_{43} = \int_0^1 (6 - 12x)(-2 + 6x) \, dx = -6.$$

$K_{34} = K_{43}$ because $K_{ij} = \int_0^1 h_i'' \, h_j'' \, dx$ is symmetric in i and j: the operator is symmetric.

c)

$$\begin{pmatrix} 12 & -6 \\ -6 & 4 \end{pmatrix} \begin{pmatrix} U_3 \\ U_4 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \Longrightarrow \quad U_3 = \frac{1}{3}, \quad U_4 = \frac{1}{2},$$
$$u_h(x) = \tfrac{1}{3}(3x^2 - 2x^3) + \tfrac{1}{2}(-x^2 + x^3) = \tfrac{1}{2}x^2 - \tfrac{1}{6}x^3.$$

Exact solution: $u'''' = 0$ gives a cubic; $u(0) = u'(0) = 0$ leaves $u = ax^2 + bx^3$; $u'''(1) = 6b = -1$ and $u''(1) = 2a + 6b = 0$ give $b = -\frac{1}{6}$, $a = \frac{1}{2}$:

$$u(x) = \tfrac{1}{2}x^2 - \tfrac{1}{6}x^3 = u_h(x), \qquad u(1) = \tfrac{1}{3}, \quad u'(1) = \tfrac{1}{2}.$$

The Galerkin solution is exact, because the exact solution is a cubic and lies in the span of the Hermite functions. The load vector is the boundary term of item 2: $F_3 = -u'''(1) = 1$, $F_4 = u''(1) = 0$.